



FACULTY/COLLEGE	College of Business and Economics
SCHOOL	School of Consumer Intelligence and Information Systems
DEPARTMENT	Applied Information Systems
CAMPUS(ES)	APB/APK
MODULE NAME	Information Systems 3A
MODULE CODE	ILS3A01
SEMESTER	First

PRACTICAL	3	WEEK OF	09 March 2026
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NUMBER OF PAGES OF QUESTION PAPER (Including cover page)	3
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INFORMATION/INSTRUCTIONS:

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- ☐ Read instructions to avoid making mistakes.
 - ☐ This is an open-book assessment.
 - ☐ Answer all the questions.
 - ☐ Upload the necessary SQL script files.
 - ☐ Read the questions carefully and answer only what is required.
 - ☐ Make sure that your submission is **an SQL file**.
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1. Write a query to display the last 5 rows of customers.

Hint: Use OFFSET & FETCH NEXT. [5]

Expected outcome:

	CustomerID	FirstName	LastName	Email	Phone	Address	City	State	PostalCode
1	15	Nombuso	Mkhize	nombuso.mkhize@example.com	0826789012	789 Oak St.	Pietermaritzburg	KwaZulu-Natal	3201
2	16	Musa	Nkosi	musa.nkosi@example.com	0737890123	567 Cedar St.	East London	Eastern Cape	5200
3	17	Lindiwe	Zulu	lindiwe.zulu@example.com	0828901234	890 Walnut St.	Bloemfontein	Free State	9301
4	18	Mandla	Mkhwanazi	mandla.mkhwanazi@example.com	0769012345	123 Oak Ave.	Polokwane	Limpopo	0700
5	19	Nkosinathi	Moyo	nkosinathi.moyo@example.com	0830123456	456 Maple St.	Klerksdorp	North West	2571

2. Write a query that will display an employee's Id, first name, last name and the number of orders each employee has served as [Orders Served]. [6]

Expected outcome:

	EmployeeID	FirstName	LastName	Orders Served
1	300	Lerato	Mthembu	7
2	301	Bongani	Nkosi	8
3	302	Sipho	Mhlongo	5
4	303	Thabo	Modise	8
5	304	Zinhle	Nkosi	7

3. Write a query to display the employee Id, first name, last name and the number of unique customers each has served as [Customers Served]. [6]

Expected outcome:

	EmployeeID	FirstName	LastName	Customers Served
1	300	Lerato	Mthembu	4
2	301	Bongani	Nkosi	4
3	302	Sipho	Mhlongo	3
4	303	Thabo	Modise	2
5	304	Zinhle	Nkosi	4

4. Write a query to display all orders along with the total number of products and the total number of items in each order. [7]

Expected outcome:

	OrderID	Number of Products	Total Items
1	400	2	8
2	403	2	3
3	404	1	10
4	410	4	16
5	411	1	5

5. A manager would like to know how many products/items were purchased from each supplier. Write a query to display the suppliers' name and the total number of products/items purchased from them. [7]

Expected outcome:

	Name	ORDERS
1	Nomalizo Ngcobo	3
2	Sandile Dlamini	1
3	Sanelisiwe Ncube	1
4	Sibusiso Nkosi	1
5	Thabo Mabuza	2
6	Thandeka Zulu	1
7	Zinhle Mkhize	1

6. Write a query to find out which supplier has the **MOST** purchased products/items.
Hint: Use a subquery to find the supplier with the maximum number of orders.
[9]

Expected outcome:

	Name	ORDERS
1	Nomalizo Ngcobo	3

7. Write a query to display all products that are **available** in the store inventory, along with their suppliers. [9]

Expected outcome:

	ProductName	Description	UnitPrice	Supplier Name
1	Camera	Mirrorless camera with 24MP sensor	12000.00	Sibusiso Nkosi
2	Glasses	Trendy sunglasses with UV protection	250.00	Sanelisiwe Ncube
3	Headphones	Wireless headphones with noise cancellation	500.00	Thandeka Zulu
4	Jeans	Blue denim jeans with slim fit	50.00	Nomalizo Ngcobo
5	Laptop	15.6-inch laptop with Intel Core i7 processor	15000.00	Nomalizo Ngcobo
6	Smartphone	Latest smartphone with 6-inch OLED display	8000.00	Sandile Dlamini
7	Sneakers	White canvas sneakers with rubber sole	100.00	Zinhle Mkhize
8	T-shirt	Cotton t-shirt with round neck	25.00	Nomalizo Ngcobo
9	Watch	Stainless steel wristwatch with leather strap	500.00	Thabo Mabuza

8. Write a query to retrieve a list of **all customers** and their orders, including customers who have never placed an order. [4]

Expected outcome:

	CustomerID	FirstName	LastName	OrderID	OrderDate	TotalAmount
1	10	Thabo	Mthembu	400	2022-01-01 00:00:00.000	250.00
2	10	Thabo	Mthembu	409	2022-10-01 00:00:00.000	1150.00
3	10	Thabo	Mthembu	415	2020-05-03 00:00:00.000	250.00
4	10	Thabo	Mthembu	424	2021-02-21 00:00:00.000	1150.00
5	10	Thabo	Mthembu	431	2022-04-25 00:00:00.000	1850.00
6	11	Zanele	Mabaso	405	2022-06-01 00:00:00.000	750.00
7	11	Zanele	Mabaso	407	2022-08-01 00:00:00.000	950.00
8	11	Zanele	Mabaso	410	2022-11-01 00:00:00.000	1250.00
9	11	Zanele	Mabaso	411	2022-12-01 00:00:00.000	1350.00
10	11	Zanele	Mabaso	420	2020-10-26 00:00:00.000	750.00
11	11	Zanele	Mabaso	422	2020-12-03 00:00:00.000	950.00
12	11	Zanele	Mabaso	425	2021-03-27 00:00:00.000	1250.00
13	11	Zanele	Mabaso	426	2021-04-30 00:00:00.000	1350.00
14	12	Bongani	Ngubane	401	2022-02-01 00:00:00.000	350.00
15	12	Bongani	Ngubane	404	2022-05-01 00:00:00.000	650.00
16	12	Bongani	Ngubane	416	2020-06-07 00:00:00.000	350.00
17	12	Bongani	Ngubane	419	2020-09-21 00:00:00.000	650.00
18	13	Nomvula	Dlamini	402	2022-03-01 00:00:00.000	450.00
19	13	Nomvula	Dlamini	403	2022-04-01 00:00:00.000	550.00
20	13	Nomvula	Dlamini	412	2023-01-01 00:00:00.000	1450.00
21	13	Nomvula	Dlamini	417	2020-07-13 00:00:00.000	450.00
22	13	Nomvula	Dlamini	418	2020-08-19 00:00:00.000	550.00
23	13	Nomvula	Dlamini	427	2021-06-03 00:00:00.000	1450.00
24	13	Nomvula	Dlamini	432	2022-05-30 00:00:00.000	1950.00
25	14	Sipho	Mahlangu	413	2023-02-01 00:00:00.000	1550.00
26	14	Sipho	Mahlangu	428	2021-07-09 00:00:00.000	1550.00
27	14	Sipho	Mahlangu	433	2022-07-04 00:00:00.000	2050.00
28	15	Nombuso	Mkhize	NULL	NULL	NULL
29	16	Musa	Nkosi	NULL	NULL	NULL
30	17	Lindiwe	Zulu	406	2022-07-01 00:00:00.000	850.00
31	17	Lindiwe	Zulu	421	2020-11-30 00:00:00.000	850.00
32	17	Lindiwe	Zulu	430	2022-03-21 00:00:00.000	1750.00
33	18	Mandla	Mkhwan...	NULL	NULL	NULL
34	19	Nkosinathi	Moyo	408	2022-09-01 00:00:00.000	1050.00
35	19	Nkosinathi	Moyo	414	2023-03-01 00:00:00.000	1650.00
36	19	Nkosinathi	Moyo	423	2021-01-15 00:00:00.000	1050.00
37	19	Nkosinathi	Moyo	429	2022-02-14 00:00:00.000	1650.00
38	19	Nkosinathi	Moyo	434	2022-08-08 00:00:00.000	2150.00

9. Write a query to retrieve a list of all orders and the **employees** who handled them, including any employees who have not processed any orders. [6]

Expected outcome:

	EmployeeID	FirstName	LastName	OrderID	OrderDate	TotalAmount
1	300	Lerato	Mthembu	404	2022-05-01 00:00:00.000	650.00
2	300	Lerato	Mthembu	405	2022-06-01 00:00:00.000	750.00
3	300	Lerato	Mthembu	414	2023-03-01 00:00:00.000	1650.00
4	300	Lerato	Mthembu	419	2020-09-21 00:00:00.000	650.00
5	300	Lerato	Mthembu	424	2021-02-21 00:00:00.000	1150.00
6	300	Lerato	Mthembu	429	2022-02-14 00:00:00.000	1650.00
7	300	Lerato	Mthembu	434	2022-08-08 00:00:00.000	2150.00
8	301	Bongani	Nkosi	430	2022-03-21 00:00:00.000	1750.00
9	301	Bongani	Nkosi	425	2021-03-27 00:00:00.000	1250.00
10	301	Bongani	Nkosi	420	2020-10-26 00:00:00.000	750.00
11	301	Bongani	Nkosi	415	2020-05-03 00:00:00.000	250.00
12	301	Bongani	Nkosi	409	2022-10-01 00:00:00.000	1150.00
13	301	Bongani	Nkosi	410	2022-11-01 00:00:00.000	1250.00
14	301	Bongani	Nkosi	400	2022-01-01 00:00:00.000	250.00
15	301	Bongani	Nkosi	401	2022-02-01 00:00:00.000	350.00
16	302	Sipho	Mhlongo	406	2022-07-01 00:00:00.000	850.00
17	302	Sipho	Mhlongo	411	2022-12-01 00:00:00.000	1350.00
18	302	Sipho	Mhlongo	416	2020-06-07 00:00:00.000	350.00
19	302	Sipho	Mhlongo	421	2020-11-30 00:00:00.000	850.00
20	302	Sipho	Mhlongo	426	2021-04-30 00:00:00.000	1350.00
21	303	Thabo	Modise	427	2021-06-03 00:00:00.000	1450.00
22	303	Thabo	Modise	422	2020-12-03 00:00:00.000	950.00
23	303	Thabo	Modise	432	2022-05-30 00:00:00.000	1950.00
24	303	Thabo	Modise	417	2020-07-13 00:00:00.000	450.00
25	303	Thabo	Modise	412	2023-01-01 00:00:00.000	1450.00
26	303	Thabo	Modise	407	2022-08-01 00:00:00.000	950.00
27	303	Thabo	Modise	402	2022-03-01 00:00:00.000	450.00
28	303	Thabo	Modise	403	2022-04-01 00:00:00.000	550.00
29	304	Zinhle	Nkosi	408	2022-09-01 00:00:00.000	1050.00
30	304	Zinhle	Nkosi	413	2023-02-01 00:00:00.000	1550.00
31	304	Zinhle	Nkosi	418	2020-08-19 00:00:00.000	550.00
32	304	Zinhle	Nkosi	433	2022-07-04 00:00:00.000	2050.00
33	304	Zinhle	Nkosi	431	2022-04-25 00:00:00.000	1850.00
34	304	Zinhle	Nkosi	423	2021-01-15 00:00:00.000	1050.00
35	304	Zinhle	Nkosi	428	2021-07-09 00:00:00.000	1550.00

10. Write a query to display all stores located in Gauteng that have at least 5 products in their inventory. [5]

Expected outcome:

	Name	Address	City	State
1	East Rand Mall	Bentel Avenue	Boksburg	Gauteng
2	Fourways Mall	William Nicol Drive	Fourways	Gauteng
3	Greenstone Shopping Centre	Stoneridge Dr & Modderfontein Rd	Edenvale	Gauteng
4	Menlyn Park Shopping Centre	Atterbury Road & Lois Avenue	Pretoria	Gauteng
5	Sandton City	5th Street & Rivonia Road	Sandton	Gauteng

Additional Tips:

- Always check the **table names** and **column names** in the database schema before writing your queries.
- Use **aliases** to make your queries more readable.
- Test your queries to ensure they return the expected results.
- Pay attention to the **marks allocated** for each question in the rubric to understand the complexity of each task.